

MAGGIE WU

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EDUCATION

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA

Candidate for Bachelor of Science in Electrical & Computer Engineering

Additional Major in Robotics & Minor in Computer Science

Anticipated Graduation: May 2028

GPA: 3.84/4.00

EXPERIENCE

SURGICAL MECHATRONICS LAB

Researcher

Carnegie Mellon University, PA

May 2025 – Present

- Developed the physical assembly of the HeartPrinter, a parallel wire robot to deliver gene therapies to the heart

ENVIRONMENTAL ENGINEERING AND SCIENCE LAB

Research Intern

Stony Brook University, NY

July 2023 – November 2023

- Characterized denitrification kinetics and greenhouse gas emission of woodchip bioreactors used in wastewater treatment
- Co-authored publication in Water Research: <https://doi.org/10.1016/j.watres.2024.122562>
- Named Regeneron STS Top 300 Scholar; Won Third Place at New York State Science & Engineering Fair

TONGE LAB

Research Intern

Stony Brook University, NY

July 2022 – August 2022

- Prepared & worked up chemical reactions to synthesize a PROTAC to selectively degrade main protease of SARS-CoV-2

PROJECT EXPERIENCE

HEARTPRINTER | CARNEGIE MELLON UNIVERSITY

May 2025 – Present

- Designed and assembled an upgraded power distribution and control unit, housed in a portable 4U rack case
- Prototyped the daughter unit (motors, load cells, EM tracker power unit), which resides next to the subxiphoid incision
- Wrote validation code in C++ for positioning the injector head relative to a plane formed by the suction bases

THE WONDER GLOVE | CARNEGIE MELLON UNIVERSITY

March 2025

- Designed and constructed a hockey glove with a flexible wrist guard to minimize wrist injuries (factor of safety > 2 for forces up to ~2500N) without inhibiting movement and a replaceable palm to improve durability and affordability
- Named MVP of a team of 4 in Rethink the Rink make-a-thon sponsored by Covestro and the Pittsburgh Penguins

AFFORDABLE STREET NAVIGATION DEVICE FOR VISUALLY-IMPAIRED PEDESTRIANS

February 2023 – June 2023

- Modified below device to detect street obstacles and vehicles with real-time audio feedback, reducing costs by over 90%
- Implemented YOLOv4-tiny neural network (trained on custom dataset), achieving 87.34% mean average precision
- Named National Finalist in JYEM Research Paper Competition (Top 5%); Presented at JYEM Expo

FACIAL RECOGNITION HAT FOR LOW-VISION USERS

December 2022 – February 2023

- Collaborated with a team of 3 and a low-vision co-designer to construct a hat integrating a camera, earbuds, and Raspberry Pi 4B to identify and verbally communicate the names of people in view
- Awarded \$125 funding in the MIT BeaverWorks CRE[AT]E Assistive Technology Challenge

LEADERSHIP EXPERIENCE, AWARDS, SKILLS

NCAA ATHLETICS – CARNEGIE MELLON VARSITY TRACK & FIELD

August 2024 – Present

- Balanced academic workload with 15 hrs/week practice and competed (throughout PA, WV, OH, NY) in hurdles & jumps

GREAT NECK SOUTH HIGH SCHOOL SCIENCE OLYMPIAD

October 2021 – June 2024

- As Treasurer, organized 10+ bake sales, raising \$1000+ over the course of a school year
- Won 18 Medals (Regional, State, National), including: 1st Place Wi-Fi Lab (constructed Double Biquad antenna), 1st Place Detector (made accurate force scale with self-made load cell), 2nd Place Trajectory (built accurate catapult for ping-pong balls)

SELECTED AWARDS: National Merit \$2500 Scholarship Recipient, Regeneron STS Top 300 Scholar

SKILLS: Programming: C++, Python, C, JavaScript, React, Next.js, CSS, x86-64 Assembly, Linux, Arduino, Raspberry Pi, Visual Studio, Git | CAD: Fusion 360, SolidWorks, Inventor | Fabrication: Soldering, Laser Cutting, 3D Printing | Data analysis: ANOVA, Excel | Currently learning: MATLAB, SML, PCB Design